UNIVERSITY OF CENTRAL PUNJAB

Faculty of Information Technology

Cyber Security

Week 7 — Cryptography Basics

What Cryptography Is • Encryption & Decryption
Symmetric vs Asymmetric • Public & Private Keys • Tools & Practice

Instructor: Hafiz Usama Ishtiaq

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QUICK RECAP

Where We Are So Far

Weeks 1–3	Cybersecurity basics, the CIA Triad, threats, and vulnerabilities.
Weeks 4–5	Service attacks, reconnaissance, and malware threats.
Week 6	Network defenses — firewalls, IDS/IPS and best practices.
This week	CRYPTOGRAPHY — the math that keeps our data secret and trustworthy.

WEEK 7 ROADMAP

What We Will Cover This Week

1

Cryptography & Encryption

What it is, key terms, and why it matters

2

Symmetric Encryption

One shared secret key — fast and simple

3

Asymmetric Encryption

Public & private key pairs — solving key sharing

4

Uses, Tools & Practice

Real-world uses, algorithms, and hands-on tools

DEFINITION

What is Cryptography?

Cryptography is the science of protecting information by turning it into a secret, unreadable form — so only the right people can read it.

In simple words: cryptography is secret writing. It scrambles a message so that anyone who intercepts it just sees nonsense.

The word: from Greek — 'krypto' (hidden) + 'graphein' (writing) = hidden writing.

A simple example

Plaintext: HELLO → shift each letter by 3 → **Ciphertext:** KHOOR

This is the famous Caesar Cipher — one of the oldest ways to hide a message.

BASIC TERMS

Cryptography Vocabulary

Plaintext	**The original, readable message.** **Example:** "Meet me at 5pm" before it is scrambled.
Ciphertext	**The scrambled, unreadable version.** **Example:** "Phhw ph dw 5sp" — nonsense to outsiders.
Encryption	**Turning plaintext into ciphertext.** **Example:** Locking the message so no one can read it.
Decryption	**Turning ciphertext back into plaintext.** **Example:** Unlocking the message with the correct key.

More Key Terms

Key	**The secret value that locks/unlocks the message.** **Example:** Like a password that controls the encryption.
Cipher / algorithm	**The method (recipe) used to encrypt.** **Example:** AES, RSA, or the simple Caesar shift.
Cryptanalysis	**The art of trying to break encryption.** **Example:** An attacker analyzing ciphertext to find the key.
Cryptology	**The whole field: cryptography + cryptanalysis.** **Example:** The study of both making and breaking codes.

IMPORTANCE

What Cryptography Protects

Cryptography supports the CIA Triad from Week 1 — plus one more goal:

Confidentiality

Keeps data secret — only authorized people can read it.

Integrity

Proves data was not changed (using hashes).

Authentication

Confirms who someone really is.

Non-repudiation

A sender can't later deny they sent it (signatures).

Secure comms

Protects messages, calls and web traffic.

Trust online

Makes safe banking, shopping and login possible.

IN DAILY LIFE

Where You Already Use Cryptography

HTTPS websites

The padlock in your browser means encrypted traffic.

WhatsApp / Signal

End-to-end encryption keeps chats private.

Online banking

Your transactions are encrypted end to end.

Wi-Fi (WPA2/3)

Your home Wi-Fi password encrypts the connection.

Passwords

Sites store hashed (not plain) passwords.

Cryptocurrency

Bitcoin and others are built on cryptography.

A QUICK HISTORY

Cryptography Through Time

Ancient	Caesar Cipher — Roman generals shifted letters to hide orders.
World War II	The Enigma machine — cracked by Alan Turing's team.
1970s	DES and public-key cryptography (RSA) were invented.
Today	AES, RSA and TLS secure the entire internet.

THE CORE IDEA

How Encryption Works

A message is locked with a key, sent safely, then unlocked with a key:



The key is everything

Even if an attacker sees the ciphertext AND knows the algorithm, they still can't read it without the correct key.

PART 2

Symmetric Encryption

One shared secret key locks and unlocks the message.

SYMMETRIC

One Key for Both Sides

In symmetric encryption, the SAME secret key is used to encrypt and to decrypt. Both sender and receiver must share it.

Analogy: like a single house key — the same key locks the door and opens it. Anyone with a copy can get in.



Examples of symmetric algorithms: AES, DES, 3DES, Blowfish.

WEIGHING IT UP

Symmetric — Strengths & Weakness

Strengths

- Very fast — good for large data
- Simple and efficient
- Uses less computing power
- Strong when the key stays secret

The big weakness

- **Key distribution problem**
- How do you share the secret key safely?
- If the key is stolen, all is exposed
- Hard to manage many keys for many people

PART 3

Asymmetric Encryption

Two keys — a public key and a private key — solve the key-sharing problem.

ASYMMETRIC

Two Keys: Public & Private

Asymmetric encryption uses a KEY PAIR: a public key (shared with everyone) and a private key (kept secret). What one locks, only the other can unlock.

Public key

Shared openly. Anyone can use it to encrypt a message TO you.

Private key

Kept 100% secret. Only YOU can use it to decrypt those messages.

Mailbox analogy

Your public key is like a mail slot — anyone can drop a letter in (encrypt). Your private key is the only key that opens the mailbox to read them (decrypt).

STEP BY STEP

Sending a Secret Message

Alice wants to send Bob a private message:

1. Bob shares his PUBLIC key with everyone.
2. Alice encrypts her message using Bob's public key.
3. The ciphertext travels safely — even if intercepted, it's unreadable.
4. Bob decrypts it using his PRIVATE key — which only he has.

Examples of asymmetric algorithms: RSA, ECC, Diffie-Hellman.

SIDE BY SIDE

Symmetric vs Asymmetric

SYMMETRIC

- One shared key
- Very fast
- Best for large amounts of data
- Hard to share the key safely
- e.g. AES, DES

ASYMMETRIC

- Key pair: public + private
- Slower
- Best for small data & key exchange
- Solves the key-sharing problem
- e.g. RSA, ECC

BEST OF BOTH

Hybrid Encryption (How HTTPS Works)

Real systems combine both: asymmetric to safely share a key, then fast symmetric encryption for the actual data.

Step 1	Your browser and the website use ASYMMETRIC keys to agree on a secret.
Step 2	They now share a symmetric 'session key' — sent safely.
Step 3	All further data uses fast SYMMETRIC encryption (AES).
Step 4	You get both safety (key exchange) and speed (bulk data).

A DIFFERENT TOOL

Hashing — One-Way Cryptography

A hash turns any data into a fixed-length 'fingerprint'. It is ONE-WAY — you cannot turn the hash back into the original.

Key point: encryption is reversible (with a key); hashing is NOT. The same input always gives the same hash.

Password storage

Sites store the hash, not your real password.

File integrity

Check a download wasn't tampered with.

Digital signatures

Hashes make signatures efficient.

Common algorithms

SHA-256 (safe); MD5 & SHA-1 (old, avoid).

PROVING IDENTITY

Digital Signatures

A digital signature proves WHO sent a message and that it was NOT changed — using the sender's private key.

How it works (reversed keys): the sender signs with their PRIVATE key; anyone can verify with the sender's PUBLIC key.

Authentication

Confirms the message really came from the sender.

Integrity

Shows the message was not altered in transit.

Non-repudiation

The sender can't deny signing it later.

Everyday use

Software updates, e-documents, certificates.

Certificates & PKI

A digital certificate proves a public key really belongs to a website. A Certificate Authority (CA) issues and vouches for it.

The padlock icon: when you see it in your browser, a trusted certificate has confirmed the site's identity and your connection is encrypted.

PKI	**Public Key Infrastructure — the whole system of keys, CAs & rules.** **Example:** The framework that makes HTTPS trust possible.
Certificate Authority	**A trusted body that issues certificates.** **Example:** Let's Encrypt, DigiCert — they vouch for websites.

THE HARD PART

Key Management

Cryptography is only as strong as how well the keys are protected. Good key management means:

Keep keys secret

Never share or expose a private key.

Use strong keys

Longer keys (e.g. AES-256) are harder to break.

Rotate keys

Change keys regularly to limit damage.

Store safely

Use secure vaults or hardware (HSM).

Back up keys

Lose the key = lose the data forever.

Revoke if leaked

Cancel and replace compromised keys fast.

Attacks on Cryptography

Brute force	**Trying every possible key until one works.** **Example:** Why long keys matter — more keys = more time to crack.
Weak algorithms	**Using outdated ciphers that are already broken.** **Example:** MD5 and DES should no longer be trusted.
Stolen keys	**Getting the key instead of breaking the math.** **Example:** Malware that steals a private key from a device.
Social / side-channel	**Tricking people or reading leaks (timing, power).** **Example:** Phishing a password, or measuring a chip's power use.

Tools to Learn & Explore

Free tools students can use to see cryptography in action.

CrypTool

Visual, learn-by-doing tool for classic & modern ciphers.

OpenSSL

Command-line toolkit to encrypt, hash and make keys.

GnuPG (GPG)

Free tool for public-key encryption and signing.

VeraCrypt

Encrypt files, folders and whole drives on your PC.

CyberChef

Browser 'Swiss-army knife' for encoding & crypto.

Hash generators

Online/offline tools to create SHA-256 hashes.

PRACTICE ACTIVITY 1

Encrypt & Decrypt a Message

A safe, fun activity in your browser — no installation needed (use CyberChef or CrypTool).

Step 1	Open CyberChef (gchq.github.io/CyberChef) in your browser.
Step 2	Type a message, apply a 'ROT13' or 'Caesar' cipher — watch it scramble.
Step 3	Reverse it to decrypt and get your message back.
Step 4	Now try 'AES Encrypt' with a key — see real modern encryption.
Step 5	Discuss: why is AES far stronger than the Caesar cipher?

PRACTICE ACTIVITY 2

See Hashing in Action

Understand why hashing is one-way and why a tiny change matters — 100% safe.

Step 1	Open a SHA-256 hash tool (in CyberChef or any hash generator).
Step 2	Hash the word 'hello' and note the long output.
Step 3	Now hash 'Hello' (capital H) — see how totally different it is.
Step 4	Try to reverse the hash back to the word — you can't (it's one-way).
Step 5	Discuss: why is this perfect for storing passwords?

QUICK REFERENCE

Key Terms at a Glance

Plaintext

The original readable message.

Ciphertext

The scrambled, unreadable message.

Encryption

Plaintext → ciphertext.

Decryption

Ciphertext → plaintext.

Symmetric

One shared key (e.g. AES).

Asymmetric

Public + private key pair (e.g. RSA).

Hashing

One-way fingerprint (e.g. SHA-256).

Digital signature

Proves sender & integrity.

RECAP

Week 7 Summary

- Cryptography is secret writing — it protects data by turning plaintext into unreadable ciphertext.
- It supports confidentiality, integrity, authentication and non-repudiation, and is used everywhere online.
- Symmetric encryption uses ONE shared key: fast, but hard to share the key safely (e.g. AES, DES).
- Asymmetric encryption uses a public + private KEY PAIR: solves key sharing but is slower (e.g. RSA, ECC).
- Hybrid systems (like HTTPS) use asymmetric to swap a key, then fast symmetric for the data.
- Hashing is one-way (SHA-256) for passwords & integrity; digital signatures prove sender and integrity.
- Practise safely with CyberChef, CrypTool, OpenSSL and GnuPG — and always protect your keys.

CHECK YOUR UNDERSTANDING

Review Questions

1. Define cryptography. Explain plaintext, ciphertext, encryption and decryption.
2. What is symmetric encryption? What is its main weakness?
3. How does asymmetric encryption work? Explain public and private keys.
4. Compare symmetric and asymmetric encryption. Give one algorithm for each.
5. What is hashing, and how is it different from encryption?
6. What is a digital signature, and what three things does it provide?

LOOKING AHEAD

The Future of Cryptography

Quantum threat

Future quantum computers may crack RSA-style keys.

Post-quantum crypto

New algorithms designed to resist quantum attacks.

Stronger keys

Moving to longer, tougher keys today.

End-to-end everywhere

More apps encrypting by default.

Privacy tech

Techniques that compute on encrypted data.

Always evolving

Cryptography keeps advancing with the threats.

WEEK 8 PREVIEW

Coming Next: Cryptography in Practice

- Encryption algorithms (AES, DES)
- Hashing algorithms (SHA, MD5) and CrypTool
- Digital signatures in depth • then Mid Term revision